Circuit Simulation Lab:26
Frequency Dividers

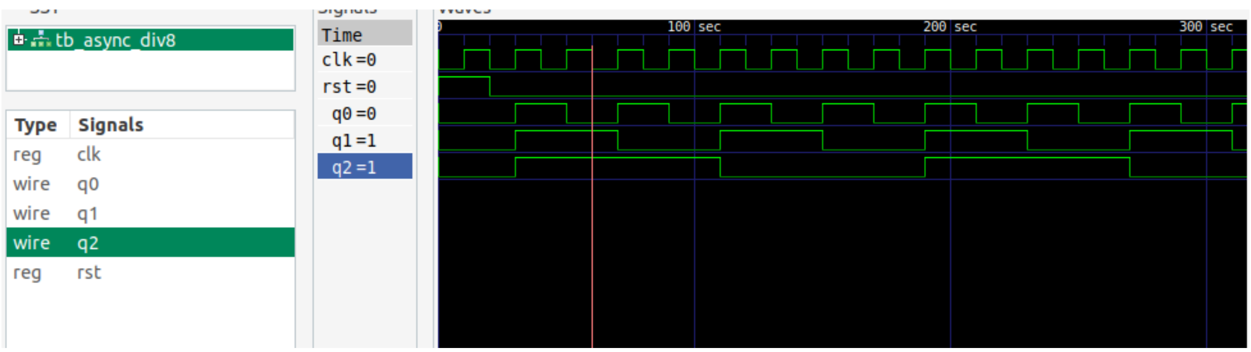
The purpose of this experiment is to introduce the design of Divide by 4, 8 Circuit.

CODE:

```
module async_div8(
input clk,
input rst,
output reg q0,
output reg q1,
output reg q2
);

always @(posedge clk or posedge rst)
begin
if(rst)
   q0 <= 0;
else
   q0 <= ~q0;
end

always @(posedge q0 or posedge rst)
begin
if(rst)
   q1 <= 0;
else
   q1 <= ~q1;
end

always @(posedge q1 or posedge rst)
begin
if(rst)
   q2 <= 0;
else
   q2 <= ~q2;
end

Endmodule
```

TESTBENCH

```
module tb_async_div8;

reg clk,rst;

wire q0,q1,q2;

async_div8 dut(
.clk(clk),
.rst(rst),
.q0(q0),
.q1(q1),
.q2(q2)
);

always #10 clk = ~clk;

initial begin

$dumpfile("async_div8.vcd");
$dumpvars(0,tb_async_div8);

clk = 0;
rst = 1;

#20;
rst = 0;

#300;

$finish;

end

initial
$monitor("t=%0t q0=%b q1=%b q2=%b",
        $time,q0,q1,q2);

endmodule
```




```
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_26$ gedit tb_async_d
iv8.v
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_26$ iverilog -o out
async_div8.v tb_async_div8.v
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_26$ vvp out
VCD info: dumpfile async_div8.vcd opened for output.
t=0 q0=0 q1=0 q2=0
t=30 q0=1 q1=1 q2=1
t=50 q0=0 q1=1 q2=1
t=70 q0=1 q1=0 q2=1
t=90 q0=0 q1=0 q2=1
t=110 q0=1 q1=1 q2=0
t=130 q0=0 q1=1 q2=0
t=150 q0=1 q1=0 q2=0
t=170 q0=0 q1=0 q2=0
t=190 q0=1 q1=1 q2=1
t=210 q0=0 q1=1 q2=1
t=230 q0=1 q1=0 q2=1
t=250 q0=0 q1=0 q2=1
```